

Robotic Systems Design Project

Aero 62520

Group 12

Project start: **Mon, 11/17/2025**

Display week: **1**

SIMPLE GANTT CHART by Vertex42.com				
https://www.vertex42.com/ExcelTemplates/simple-gantt-chart.html				
TASK	RESPONSIBILITY	PROGRESS	START	END
Preliminary Design Review (PDR)				
Introduction	Qikai Sun	100%	17/11/25	09/12/25
System	Zhikun Peng	100%	17/11/25	09/12/25
Mechanical Design	Nicholas Lesseps	100%	17/11/25	09/12/25
Electrical Design	Anitra Bag	100%	17/11/25	09/12/25
Software Design	Shaokong Hong	100%	17/11/25	09/12/25
Analysis	All (Individual Sections)	100%	17/11/25	09/12/25
Project Plan	All (Individual Sections)	100%	17/11/25	09/12/25
Review and LaTeX	All (QC Process)	100%	07/12/25	11/12/25
Preliminary Design Review Deadline			12/12/25	
Technical Competency Assessment			All	
			17/12/25	
Manipulation				
Arm connection to Nuc	Zhikun Peng	100%	17/11/25	19/11/25
Preliminary Testing	Zhikun Peng, Anitra Bag	100%	19/11/25	09/12/25
Arm kinematics for block gripping &	Anitra Bag, Nicholas Lesseps	100%	04/12/25	19/12/25
Arm autonomous gripping	Zhikun Peng, Nicholas Lessr	100%	02/02/26	16/02/26
Arm autonomous deposition	Zhikun Peng, Nicholas Lessr	100%	16/02/26	23/02/26
Arm gripping & deposition (rover mo	Zhikun Peng, Nicholas Lessr	100%	23/02/26	02/03/26
Arm retain block in transit (rover mo	Zhikun Peng, Nicholas Lessr	100%	02/03/26	09/03/26
Sucessful Manipulation Video Evidence			09/03/26	
Vision				
Camera Setup & Driver Integration	Anitra Bag	100%	02/12/25	07/12/25
Data Set Colletion	Qikai Sun, Anitra Bag	100%	01/12/25	19/12/25
Object Detection Algorithm	Qikai Sun, Anitra Bag	100%	08/12/25	08/02/26
Camera Calibration	Qikai Sun, Anitra Bag	100%	13/12/25	19/12/25
Camera-Gripper Calibration	Qikai Sun, Anitra Bag	100%	08/02/26	22/02/26
Coordinate Transformation Testing	Qikai Sun, Anitra Bag	100%	22/02/26	08/03/26
Sucessful Vision Video Evidence			08/03/26	
Navigation				
Run Nav2 bringup on NUC, ensure dependencies and ROS2 packages are correctly installed	Shaokong Hong	100%	17/11/25	23/11/25
Find exploration strategies (frontier-based, coverage-based), review mxelore-ros2 suitability,	Shaokong Hong	100%	23/11/25	30/11/25
Prepare integration tests for baseline navigation pipeline (Nav2)	Shaokong Hong	100%	30/11/25	07/12/25
Real-robot testing to verify functional feasibility of core navigation pipeline on hardware.	Shaokong Hong	100%	07/12/25	19/12/25
Simulation testing exploration strategies using preliminary sensor configuration (including inverted	Shaokong Hong	100%	02/02/26	12/02/26
Modify existing explore node to expose an Action interface for improved orchestration and interrupt	Shaokong Hong	100%	12/02/26	19/02/26
Integrate the Action-enabled explore node with the bringup node and conduct full simulation testing.	Shaokong Hong	100%	19/02/26	26/02/26
Test external interrupt mechanisms, including action cancellation and service-based overrides.	Shaokong Hong	50%	26/02/26	05/03/26
Sucessful Navigation Video Evidence			05/03/26	
Integratgion				
Full System Assembly (mechanical)	Nicholas Lesseps	100%	02/02/26	09/02/26
Power Systems Testing and Integrs	Anitra Bag	0%	23/03/26	30/03/26
Arm-Camera-NUC Integration	Zhikun Peng, Qikai Sun	80%	16/02/26	30/03/26
Arm-Camera-NUC-Rover Integration	Shaokong Hong, Qikai Sun	60%	02/03/26	30/03/26
Full system Integration	All (Individual Sections)	0%	20/03/26	03/04/26
Final Design Review Deadline			All	
			20/03/26	
Peer Review 3 Deadline			All	
			20/03/26	
Full System Testing	All (Individual Sections)	0%	03/04/26	04/05/26
Final Checks	All (Individual Sections)	0%	04/05/26	08/05/26
Peer Review 4 Deadline			All	
			08/05/26	
Technical Demonstration Date			All	
			08/05/26	
Technical Evaluation Video Deadline			All	
			08/05/26	

